

Gem5 Cache & NVM Emulation Assignment Report - Group 40

1. Submitted Files Overview

Part A: Cache Analysis Source Files

- `src/mem/ruby/structures/CacheMemory.hh`
 - **Purpose:** Added member variables `m_window_hits` and `m_window_misses` to track performance within a specific timing window. Added the function declaration for `regStats()`.
- `src/mem/ruby/structures/CacheMemory.cc`
 - **Purpose:** Implemented the windowed counting logic. Modified `profileDemandHit()` and `profileDemandMiss()` to increment counters only if the current tick is between 2,750,000,000 and 3,250,000,000 (Group 40 window). Implemented `regStats()` to print these results to the console at simulation end.

Part B: NVM Emulation Source Files

- `src/mem/DRAMInterface.py`
- `src/python/gem5/components/memory/dram_interfaces/ddr3.py`
 - **Purpose:** Defined the `NVM_PCM` memory class inheriting from `DDR3_2133_8x8`. Scaled the timing parameters according to the PCM profile requirements: 3x multiplier on read latencies (`tRCD`, `tCL`, `tRP`, `tRAS`) and 10x multiplier on write latencies (`tWR`, `tRTP`).

Simulation Logs

- `baseline_dram_log.txt`: Complete output of the simulation using standard DDR3 memory.
- `nvm_pcm_log.txt`: Complete output of the simulation using the emulated PCM memory.

2. Part A: Cache Statistics (L1-I Cache)

Results extracted from the simulation measurement window [5.5M - 6.5M cycles]:

Metric	Value
Window Hits	328,607
Window Misses	1,294
Miss Rate	0.3922%

3. Part B: Execution Cycles Calculation

The total execution cycles are calculated by dividing the final simulation ticks by the conversion factor (1 clock cycle = 500 ticks).

DRAM (Baseline)

- **Final Tick:** 21,608,219,500
- **Calculation:** 21,608,219,500 / 500
- **Total Cycles:** 43,216,439

NVM (PCM Emulation)

- **Final Tick:** 23,261,520,000
- **Calculation:** 23,261,520,000 / 500
- **Total Cycles:** 46,523,040

4. Conclusion

The transition from DDR3 DRAM to Phase Change Memory (PCM) resulted in an increase of **3,306,601 cycles**, representing a performance degradation of approximately **7.65%**.

This difference occurs because PCM has significantly higher read and write latencies compared to DDR3. When the processor encounters an L2 cache miss and must fetch data from main memory, it remains in a "stalled" state for a longer duration under the PCM configuration. Despite the L1-I cache maintaining a relatively low miss rate (~0.39%), the memory-intensive nature of the `505.mcf_r` workload ensures that these cumulative latency penalties have a measurable impact on the overall execution time.